

WJEC GCE BIOLOGY - HUMAN BIOLOGY
SUMMER 2017
UNIT 4 MARK SCHEME

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		(Genetically) Identical / clone/ the same (1) Mitosis (1)		2		2		
		(ii)		Stigma above anthers/ anthers below stigma/ carpels above stamen/ reference to female parts above male parts/ ORA (1) NOT they are different heights		1		1		
		(iii)		Prevent inbreeding / (Increase) <u>genetic</u> {variation/ diversity}/Prevent homozygosity/ description of homozygosity/ larger gene pool/ ORA (1) increase chances of survival (of plant species)/ selective advantage/ adapt to changing environment/ maintain vigour (1)		2		2		
	(b)	(i)		Gamete 10 Endosperm 30 Petal 20 3 correct = 2 marks, 2 correct = 1 mark		2		2		
		(ii)		A. Chromosome number doubled / $3n$ to $6n$ / triploid to hexaploid/ even number of <u>each</u> chromosome (1) NOT returned to diploid B. ref to endomitosis/ cell does not divide/ non disjunction/ no cytokinesis (1) C. Homologous pairs/ bivalents form (1) D. Prophase 1 (1) E. Meiosis (1) F. (viable) gametes can be produced (1)	4	1		5		